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Body fluids

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In the average young adult male , 18% of the body - weight is protein & related substances , 7% is mineral . & 15% is fat

. The remaining 60% is water-

-Total body water is comprised of extracellular & intracellular fluid.

The intracellular fluid accounts for about 40% of- .body weight

The extracellular fluid (about 20% of total body- weight)can be subdivided into two main sub -: compartments

The **plasma** , which makes up about 5% of body- .weight

And the **interstitial fluid** which lies between the tissue- .cells which makes up about 15% of body weight

The percentage of total body water is **greater** in newborns, and **lower** in adult females , elderly persons , or adults with a large amount of . adipose tissue

Because the plasma & interstitial fluids are-separated only by highly permeable capillary membranes , their ionic compositions are similar & they are often considered together as one . large compartment of homogeneous fluid

-The most important **difference** between plasma & interstitial fluid is the higher concentration of protein in the plasma , which exists because the capillaries have a low permeability to the plasma proteins .

- -**Extracellular fluid** (the major cation is Na^+ and the major anions are Cl^- & HCO_3^-).
- -**Intracellular fluid** (The major cations are K^+ & Mg^{+2} & the major anions are protein & organic phosphates such as ATP, ADP & AMP).

Differences between ECF & ICF

ECF

Cations:

Na^+ (142_{mmol/L})

K^+ (4.2)

Mg^{2+} (0.8)

Anions:

Cl^- (108)

HCO_3^- (24)

Nutrients:

O_2 , glucose, fatty acids, & amino acids.

Wastes:

CO_2 , Urea, uric acid, excess water, & ions.

ICF

Cations:

Na^+ (14)

K^+ (140)

Mg^{2+} (20)

Anions:

Cl^- (4)

HCO_3^- (10)

Phosphate ions

Nutrients:

High concentrations of proteins.

COMPOSITION OF BODY FLUIDS

CATIONS (mmol/l)	Plasma	Interstitial	Intracellular
Na	142	139	14
K	4.2	4.0	140
Ca	1.3	1.2	0
Mg	0.8	0.7	20
ANIONS (mmol/l)			
Cl	108	108	4.0
HCO₃	24.0	28.3	10
Protein	1.2	0.2	4.0
HPO₄	2.0	2.0	11

-:Nature of body fluids

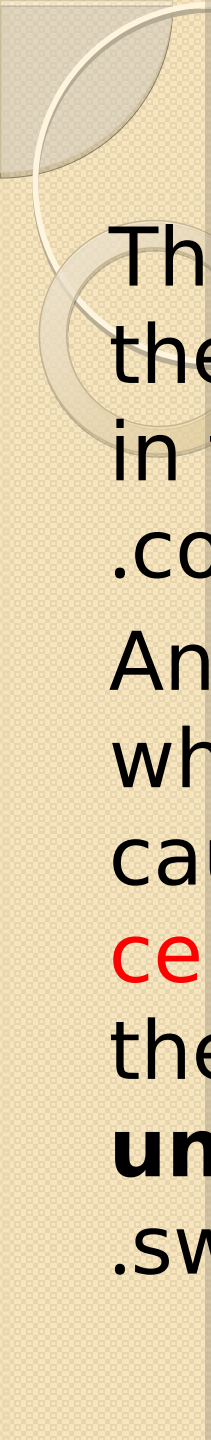
A solution is said to be isotonic if no osmotic force develops across the cell membrane

This means that an isotonic solution has the same osmolarity as the cell & that the cells will not shrink or swell if placed in this solution

Example of isotonic solutions are: -(0.9% NaCl & a .5% glucose solution)

A solution is said to be hypertonic when it contains a higher osmotic concentration of substances than does the cell

In this case , osmotic force develops that causes water to flow out the cell into the solution , thereby greatly concentrating intracellular fluid & shrinking the cell



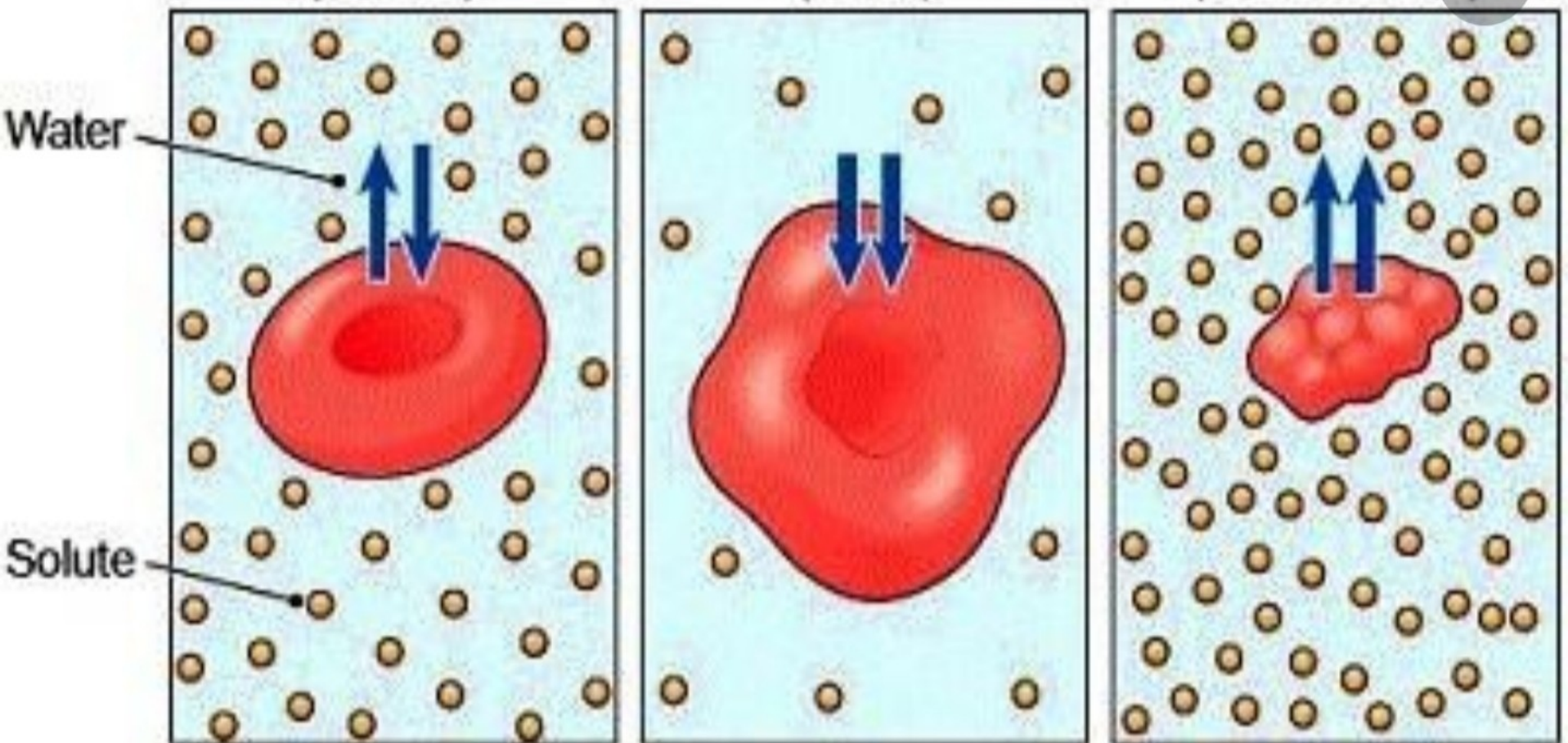
The solution is said to be hypotonic if-
the osmotic concentration of substances
in the solution is **less** than their
.concentration in the cell

An osmotic force develops immediately-
when the cell is exposed to the solution ,
causes water to flow by osmosis **into the
cell** until the intracellular fluid has about
the same concentration as the fluid , or
until the cell bursts from excessive
.swelling

Isotonic solution
(normal)

Hypotonic solution
(dilute)

Hypertonic solution
(concentrated)



A Normal red
blood cell

B Swollen red
blood cell

C Shrunk (crenated)
red blood cell

Direction of osmotic
water movement

-: Transport across cell membranes-

1-Simple diffusion

2-Carrier -mediated transport

❖-Facilitated diffusion

❖-Primary active transport

❖-Secondary active transport

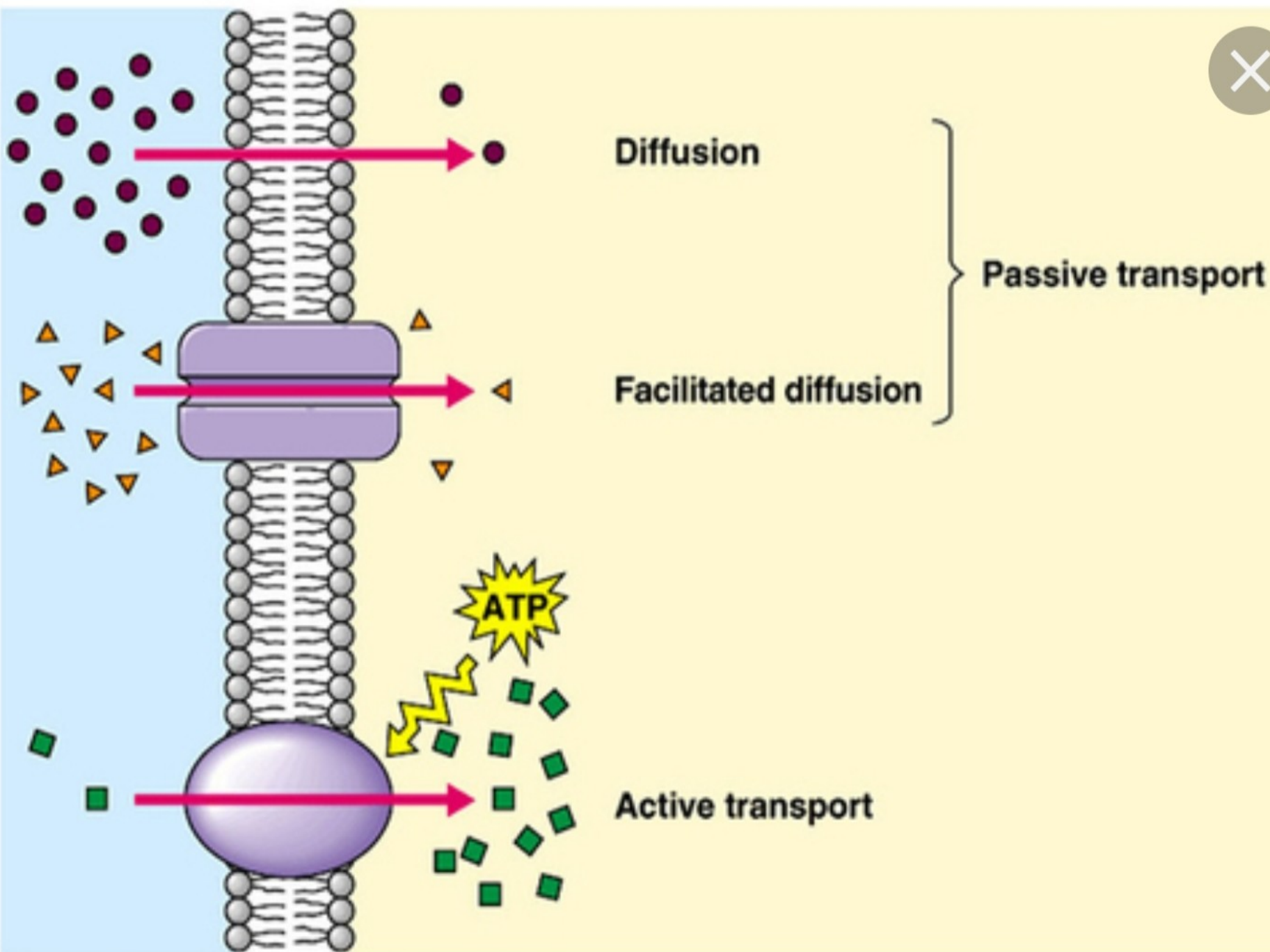
3- Osmosis

-: Simple diffusion-

- occurs down an electrochemical gradient, "downhill"-which is molecules migrate from a region of high concentration to one lower concentration .
- -This form of transport is **not carrier** mediated .
- -**Not** require metabolic energy , therefore is passive .
- -The rate of diffusion across the cell membrane is directly related to :-
- -(1):-The electrical potential & chemical concentration differences across the membrane
- -(2):-The surface area of the membrane .
- -(3):- The permeability of the membrane for the solute, which is inversely related to the size of the solute & the membrane thickness.

- :Facilitated diffusion-

- Occurs down an electrochemical gradient (“**downhill**”), similar to simple diffusion.
- Does **not** require metabolic energy & therefore is passive.
- Is **more** rapid than simple diffusion.
- Is **carrier** – mediated.
- Ex. Glucose transport in muscle & adipose cells is “downhill” is carrier – mediated.
- In diabetes mellitus , glucose uptake & use by muscle & adipose cells are impaired because the carriers facilitated diffusion of glucose require insulin .



-:Primary active transport-

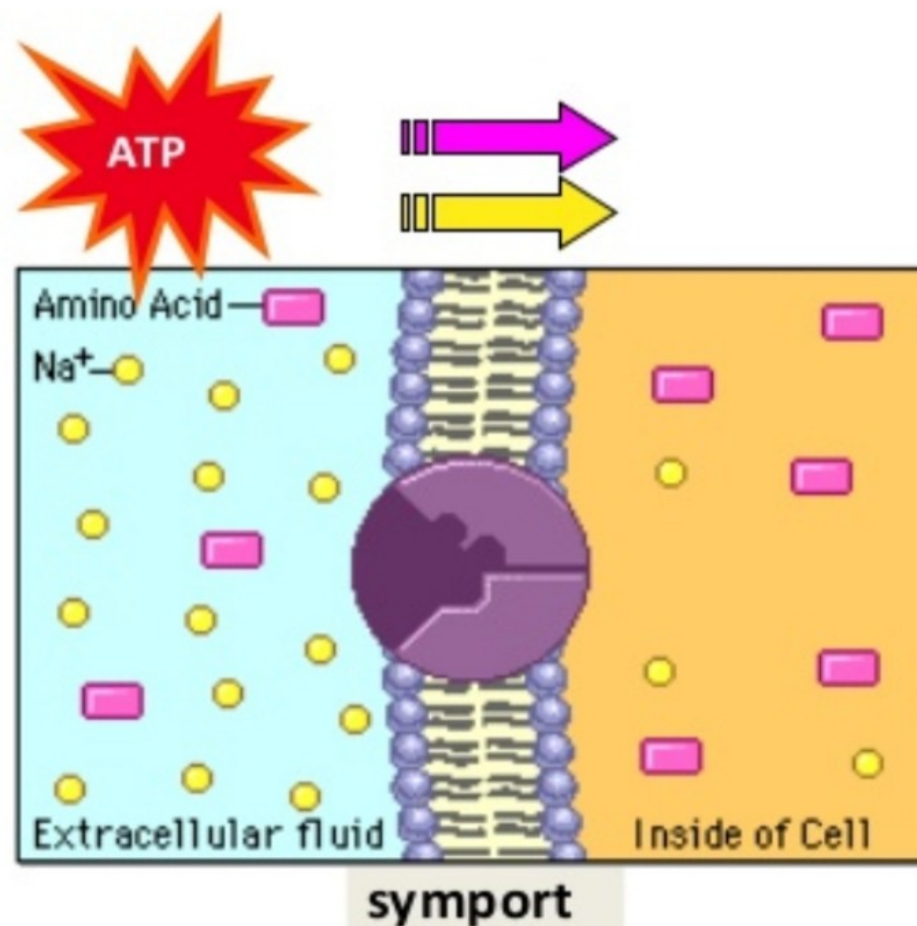
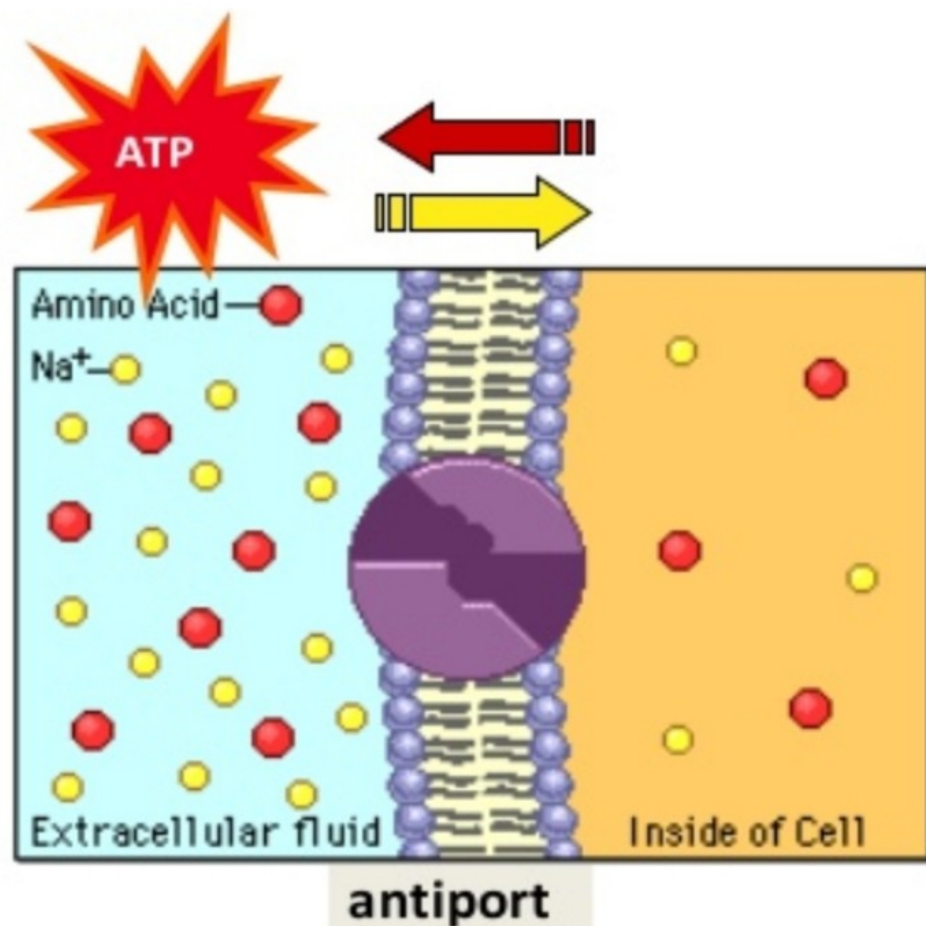
- Occurs against an electrochemical gradient (“**uphill**”).
- Requires direct input of metabolic energy in form of ATP & is active.
- Is **carrier** – mediated.
- Ex: Na^+ , K^+ - ATPase (or Na^+ - K^+ pump) in the cell membrane transports Na^+ from intracellular to extracellular fluid & K^+ from extracellular to intracellular fluid ; It maintains low intracellular Na^+ & high intracellular K^+ .
- Both Na^+ & K^+ are transported against their electrochemical gradients .
- Energy is provided from ATP.

- :Secondary active transport

- In this transport ; transport of **two or more** solutes is coupled. One of the solutes (usually Na^+) is transported “downhill” & provides energy for the “uphill” transport of the other solute .
- Metabolic **energy** is **not provided directly** , but indirectly from the Na^+ gradient , which is maintained across cell membranes . Thus , inhibition of Na^+ , K^+ - ATPase will decrease transport of Na^+ out of the cell , decrease the transmembrane Na^+ gradient , & eventually inhibit secondary active transport .
- If the solutes move in the same direction across the cell membrane , it is called **cotransport (symport)**, Ex: (Na^+ - glucose cotransport in the small intestine).
- If the solutes move in opposite directions across the cell membranes , it is called **counter transport (antiport)**, exchange, Ex: (Na^+ – Ca^+ exchange or Na^+ – H^+ exchange) .

Active transport

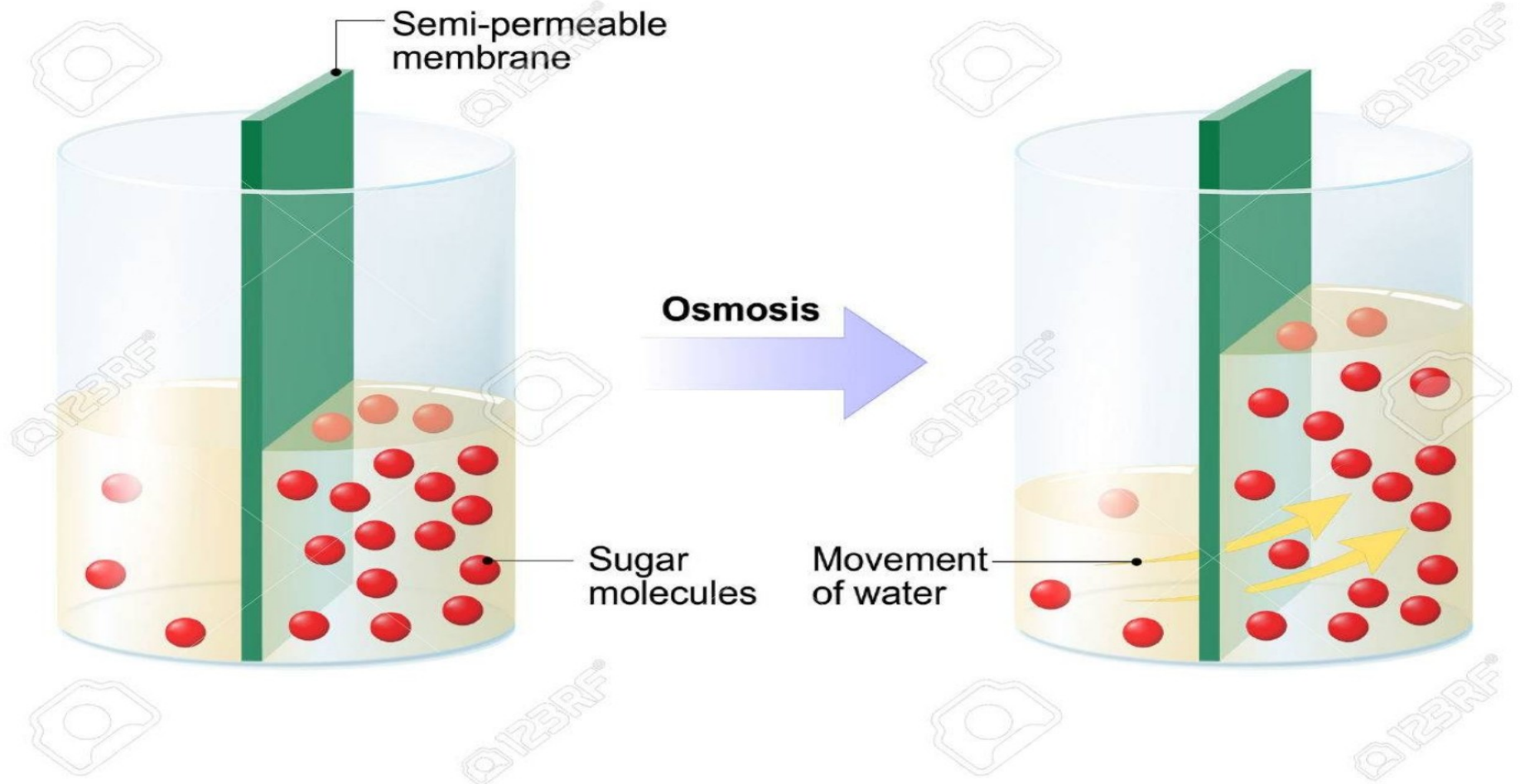
- Many models & mechanisms



-:Osmosis

- -Osmosis is the **flow** of water across a **semipermeable** membrane from a solution with low solute concentration to a solution with high solute concentration.
- -Cell membranes in the body are highly permeable to water & whenever there is a higher concentration of solute on one side of the membrane , water rapidly diffuses across the membrane toward the region of higher solute concentration .

OSMOSIS



Fluid Filtration Across Capillaries

- Fluid pressure and colloid osmotic pressure forces operate at the capillary membrane, tending to move fluid either outward or inward through the membrane pores.
- **The hydrostatic pressure** in the capillaries tends to force fluid and its dissolved substances through the capillary pores into the interstitial spaces.
- Conversely, **osmotic pressure** caused by the plasma proteins (called *colloid osmotic pressure*) tends to cause fluid movement by osmosis from the interstitial spaces into the blood. This osmotic pressure exerted by the plasma proteins normally prevents significant loss of fluid volume from the blood into the interstitial spaces.

Plasma Colloid Osmotic Pressure

- **Proteins in the Plasma Cause Colloid Osmotic Pressure.**
- Normally only those molecules or ions that fail to pass through the pores of a semipermeable membrane exert osmotic pressure. Because **the proteins are the only dissolved constituents in the plasma and interstitial fluids that do not readily pass** through the capillary pores, it is the proteins of the plasma and interstitial fluids that are responsible for the osmotic pressures on the two sides of the capillary membrane.
- To distinguish this osmotic pressure from that which occurs at the cell membrane, it is called either *colloid osmotic pressure* or *oncotic pressure*.